1. **Agentic AI:** Agentic AI is an approach where multiple agents can work together and handle different tasks to achieve one main goal. I implemented this concept in my Agentic Chatbot project using LangGraph with nodes and edges. I created different nodes for answering user questions, tool calling and fetching weekly AI news. Each node has its own responsibility and works as part of the workflow. This approach helps us build AI applications where different agents can handle tasks independently and work together to complete the overall process.

2. **Workflow AI**: Workflow AI connects different tasks and tools to complete a process automatically. For example a workflow can collect data and send it to an AI model and then store the result in a database and notify the user. Each step depends on the previous step and the workflow manages the complete process.

3. **AI copilots / assistants**: AI copilots or assistants help users complete their tasks. They can understand user questions and provide suggestions and generate content or help with coding and data analysis. They support users in their work instead of completely replacing them.

4. **Standalone AI model vs AI integrated into a product workflow**: A standalone AI model takes input and generates a prediction or output. AI integrated into a product workflow connects the model with the application and database and other tools. It can use the model output to perform further actions and help users complete their tasks inside the product.

5. **GenAI, LLMs, RAG and AI Agents** : I have worked on GenAI projects using Gemini API and Hugging Face. I developed DocPilot AI using RAG where I worked on document processing chunking embeddings and vector retrieval to generate answers based on the given documents. I also developed a real time speech to speech sales automation system using Gemini Live API. I have also explored AI agents and how they can handle tasks through different workflows.

6. **Workflow Automation:** I have worked on workflow automation using n8n where I designed an AI-based email automation workflow. When a new email arrives the workflow checks the calendar to identify any upcoming meetings or interview schedules and prepares a suitable reply based on that information. Instead of sending the email automatically I configured it to save the reply as a draft for human review and approval. I followed this approach because once an email is sent we cannot edit it so human verification is important before sending. This helps reduce manual work while keeping control over the final response.

7. **AI Integration into a Web Platform or Product** : I have experience integrating AI features into applications using Python and Streamlit. In DocPilot AI I connected document processing and RAG functionality with the application so users can upload documents and ask questions. I have also worked on integrating Gemini API to build real time conversational AI features.

Task 3 – AI Use Case Evaluation

1. AI-Assisted Input Validation and Missing Parameter Detection

Problem it solves:
While working with engineering simulations engineers may forget to enter some required parameters or provide incorrect values. This can lead to errors during calculations. An AI-assisted validation feature can identify missing parameters and values outside the allowed range before running the simulation.

Data required:
Engineering input parameters such as TDP, air velocity, TIM conductivity, heat-sink dimensions, material properties and their allowed ranges.

AI technology:
Workflow AI and rule-based validation. ML can also be used to identify unusual input combinations based on previous engineering data.

Validation:
Test the feature with correct inputs, missing values and incorrect values. Check whether it identifies the issues correctly and prevents invalid inputs from reaching the simulation.

2. ML Surrogate Model for Fast Thermal Prediction

Problem it solves:
Running engineering simulations repeatedly for different input combinations can take time. An

ML surrogate model can predict thermal resistance and junction temperature based on the given parameters and help engineers evaluate different designs quickly.

Data required:
Simulation datasets containing TDP, air velocity, TIM conductivity, thermal resistance and junction temperature.

AI technology:
Machine Learning using models such as Random Forest and Gradient Boosting.

Validation:
Compare the predicted results with the original physics-based simulation using MAE, RMSE and R^2. Also test the model with new input combinations that were not used during training.

3. AI-Based Thermal Design Recommendations

Problem it solves:
Engineers may need to try different heat-sink parameters to achieve better cooling performance. Finding suitable combinations manually can take time. An AI-based recommendation feature can suggest changes in airflow, TIM conductivity and heat-sink design parameters to reduce thermal resistance and junction temperature.

Data required:
Previous simulation results, heat-sink dimensions, material properties, TDP, air velocity, TIM conductivity and temperature limits.

AI technology:
Machine Learning and optimization algorithms. The ML model predicts thermal performance and an optimization algorithm searches for suitable design combinations.

Validation:
Compare the recommended design with the original design using the physics-based simulation. Check whether the recommendations improve thermal performance and satisfy the specified temperature limits.

4. AI Engineering Document Assistant

Problem it solves:
Engineers work with technical documents and simulation reports. Finding specific information from multiple documents manually takes time. An AI assistant can help engineers upload documents and ask questions to get answers based on the available information.

Data required:
Engineering PDFs, technical documentation, product specifications, simulation reports and design manuals.

AI technology:
GenAI, LLMs and RAG. Documents can be processed into chunks and converted into embeddings. Vector retrieval can find relevant information and provide it as context to the LLM to generate answers.

Validation:
Compare the generated answers with the original documents. Check whether the answers are accurate and relevant and whether the system avoids generating information that is not available in the documents.

5. Automated Thermal Result Summarization and Report Generation

Problem it solves:
After running simulations engineers need to understand the results and prepare reports. Manually reviewing multiple outputs and preparing reports takes time. An AI feature can summarize simulation results and generate reports containing important parameters, thermal resistance, junction temperature and possible issues.

Data required:
Simulation inputs, calculated thermal results, ML predictions, comparison metrics, plots and previous engineering reports.

AI technology:
GenAI and Workflow AI. The workflow collects simulation results and passes the information to an LLM to generate a structured summary and report.

Validation:
Compare the generated report with the actual simulation outputs. Verify that all numerical values are correct and that the report does not introduce unsupported conclusions. Engineers should review the report before final approval.

Task 4: Motivation and Ownership

1. Why do you want to join Expert Thermal / XThermal?

I want to join Expert Thermal because I see this as a good opportunity to work on real-world problems where AI and ML can be applied beyond software applications. I am interested in understanding how AI can be used in thermal engineering and scientific domains. I also want to learn from experienced people and contribute to building practical solutions.

2. What are you most interested in building in AI/ML?

I am most interested in building AI applications that solve real-world problems. I like working with LLMs, RAG, AI agents and automation workflows. At the same time I want to explore how ML models can be used with real-world data to make predictions and support decision making.

3. Are you more interested in ML models, GenAI workflows, physics-aware AI, or product AI integration?

I am interested in all these areas but I am particularly interested in GenAI workflows and physics-aware AI. I have already worked on GenAI projects like RAG applications and AI agents. Through this assignment I am also getting interested in how we can combine physics-based calculations with ML models to build faster and more useful solutions.

4. How do you handle ambiguous problems where requirements are not fully defined?

First I try to understand the actual problem and what result we are expecting. If anything is unclear I break the problem into smaller parts and identify what information is missing. Then I start with a basic approach and test it with available data. Based on the results and feedback I make improvements instead of waiting for everything to be perfectly defined.

5. Are you comfortable working in a fast-moving startup environment?

Yes I am comfortable with it. I understand that in a startup we may need to learn new technologies quickly and handle different responsibilities. I am ready to take challenges, adapt to changes and work on tasks where I may not have previous experience.

6. How do you think you can grow within our team?

I want to grow by working on real projects and understanding how things are developed and used in production. Initially I want to improve my technical knowledge and learn from the team. As I gain experience I want to take responsibility for complete tasks and eventually contribute to designing and developing solutions independently.

7. How would you bring new ideas, create solutions, and take on greater responsibility over time?

I would first understand the existing workflow and identify where improvements can be made. Whenever I get an idea I would try to build a small prototype and check whether it actually solves the problem. I believe in taking ownership of the work assigned to me rather than just completing the task. As I gain more knowledge I would like to suggest improvements, explore new tools and take responsibility for bigger parts of the project.